

C2_03 Gate B Protected Review - Astra RC4

Clinic Encounters & Billing. Fresh remediation reassessment only; open after a saved Gate B attempt.

1. Objective controls

Control	Expected	Why it matters
Fact rows	Encounters 90 Billing 60	Different business facts; only 60 encounters have a billing row.
Provider key	9 raw / 8 unique duplicate PR03	One-side Provider key must be unique before relationship.
Unknown provider	PR99 on encounter E088 charge base 2,236	The unknown row is Cancelled, so a completed-only KPI can look unchanged while the model is still invalid.
Encounter status	60 Completed / 30 Cancelled	Useful independent population control.
Money controls	Encounter ChargeBase 122,665 Billing 88,257.60	Not the same business measure/population; do not force them to reconcile directly.
Date-role contrast	Visit month rows May/Jun/Jul = 41/40/9; Discharge = 40/39/11	Visit vs discharge reporting legitimately shifts period counts.
Targets	9 Specialty-Month rows target units 1152	Higher-grain target must not repeat across encounter rows.

2. Expected reasoning path

FactEncounter is one encounter and FactBilling is one billing record. The 60 billing rows reference 60 unique encounter IDs; absence of duplicate billing does not make Billing a dimension.

PR03 is duplicated but its attributes match in this fixture. Prove that before deterministic dedupe; never use many-to-many simply because the raw one-side is dirty.

PR99 occurs only on a cancelled E088. If a learner validates only Completed counts, the defect can hide. A professional QA pack checks all source rows, orphan counts and charge controls as well as decision KPIs.

VisitDate and DischargeDate answer different questions. The period counts differ at boundaries, so the default role must be explicit and the alternate path tested.

Specialty-Month targets require a conformed Specialty/Date route. Encounter ChargeBase and Billing BillAmount are different measures/populations; a mismatch between 122,665 and 88,257.60 is not itself an error.

3. Critical failures

- Calls billing a dimension or joins facts in a way that changes encounter grain.
- Ignores PR99 because completed-only metrics happen not to change.
- Forces ChargeBase and BillAmount to equal instead of defining their different populations.
- Uses VisitDate and DischargeDate interchangeably.
- Repeats Specialty-Month target for each encounter.
- Signs off without exception counts, filter tests and a release/block decision.

4. Competency floors

Competency	Required proof
Model contract / grain	Every fact and higher-grain table has an explicit row meaning; no mixed-grain shortcut.
Keys / star design	One-side uniqueness and orphan handling are evidenced before relationships; no arbitrary dedupe.
Relationships / date roles	Cardinality, active state and direction documented; tiny filter tests prove both required date roles.
Power Query / reconciliation	Traceable source-staging-model-ready path plus independent rows/keys/totals/exceptions.
Performance / handoff	Correctness first; runtime performance evidence only when actually observed; RELEASE/BLOCK and repair path are explicit.

Changed repair proof

Close this review and prove the repair on a changed provider/date/target fixture. Gate C remains a different full logistics reassessment and opens only when explicitly assigned.